Radial Flow Interpretation of the Gravitational Invariant $E = \omega^2 R^3$:

A Hydrodynamic Model for Free-Fall Acceleration Without Mass

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Abstract

Papers I and II of this series established that the orbital invariant $E = \omega^2 R^3$ is a mass-free operational constant verified across twenty-nine orders of magnitude in E, and that $g = E/R^2$ reproduces free-fall acceleration for all Solar System bodies to within 0.22%. The present paper (Paper III) addresses the physical mechanism underlying this result. We propose a model in which a non-material radial flow field converges toward the centre of a gravitational system, carrying E as a kinematic property of the field. The characteristic velocity of this flow at radius R satisfies $E = v_{\text{flow}}^2 \cdot R^3$, in formal analogy with the Rayleigh–Plesset equation for spherical cavity collapse in incompressible fluid. However, a material body in this flow is not a collapsing cavity wall (which yields a factor 3/2 in the equation of motion) but a test particle experiencing the local pressure gradient of the flow, which yields $g = E/R^2$ exactly without any additional free parameters. We present the derivation in full mathematical detail (Appendices A–C), explain the physical origin of the distinction between the Rayleigh case and the present case, identify the specific mathematical step at which the factor 3/2 appears and disappears, and verify free-fall kinematics by fourth-order Runge–Kutta (RK4) numerical integration against classical mechanics for five bodies at five heights. Agreement is within 0.48% for all tested cases; discrepancies are fully explained by the constant-g approximation used in the classical reference. The model introduces no free parameters, proposes no modification to Newtonian mechanics, and provides a kinematic-geometric derivation of gravitational acceleration that does not require mass as a primary concept.

Keywords: gravitation; hydrodynamic analogy; Rayleigh–Plesset equation; radial flow; free-fall kinematics; pressure gradient; orbital mechanics; non-material field.

Contents

1 Introduction and Physical Motivation **3**

2 Related Work: Hydrodynamic and Field Analogies for Gravity **3**
2.1 The Unruh Acoustic Analogy . . . 3
2.2 Gordon's Optical Metric . . . 4
2.3 Verlinde's Entropic Gravity . . . 4
2.4 The Rayleigh–Plesset Equation in Astrophysics . . . 4
2.5 Non-Material Flows and the Aether Debate . . . 4

3 Two Physically Distinct Objects **5**
3.1 The Material Body . . . 5
3.2 The Non-Material Radial Flow . . . 5

4 The Rayleigh–Plesset Equation: Full Derivation and the Factor 3/2 **6**
4.1 Setup: Spherical Cavity Collapse . . . 6
4.2 Velocity Field and Kinetic Energy . . . 6
4.3 Energy Conservation and the Wall Equation of Motion . . . 6
4.4 First Integral and Formal Identity with E . . . 7

5 The Critical Physical Distinction: Wall vs. Test Particle **7**
5.1 What Changes When the Body Is Not the Wall . . . 7
5.2 Derivation of $g = E/R^2$ from the Flow Pressure Gradient . . . 8

6 A Key Physical Insight: v_{flow} at the Surface Equals the First Cosmic Velocity **9**

7 Numerical Verification of Free-Fall Kinematics **10**
7.1 Method . . . 10
7.2 Results . . . 10
7.3 Convergence Check . . . 11

8 Discussion **11**
8.1 What the Model Claims and Does Not Claim . . . 11
8.2 Scope of the Interpretation . . . 11
8.3 The Three-Paper Logical Structure . . . 11

9 Conclusions **12**

Acknowledgments **12**

Statements and Declarations **12**

Data Availability Statement **12**

References **12**

Appendix A Step-by-Step Derivation of the Rayleigh–Plesset Equation and the Factor 3/2 **14**

Appendix B Derivation of $g = E/R^2$ from Flow Pressure Gradient **15**

Appendix C Surface Flow Velocity and First Cosmic Velocity **16**

1 Introduction and Physical Motivation

Papers I [1] and II [2] of this series established two empirical results. Paper I showed that the expression $E \equiv \omega^2 R^3$, derivable from Kepler's Third Law without reference to mass or the gravitational constant G, constitutes a self-sufficient operational constant of any gravitational system: specifically, that $g = E/R^2$ reproduces official surface-gravity values for all eight Solar System planets with maximum error 0.22%. Paper II extended this verification across twenty-nine orders of magnitude in E, from the Bohr orbit of the hydrogen atom to the Sun's galactic orbit, and confirmed E-invariance using multiple independent orbital probes for each system.

These results raise a question that both papers deliberately left open: what physical mechanism produces the exact result $g = E/R^2$? The question is more specific than "what is the cause of gravity" in a metaphysical sense: given the established kinematic fact that $E = \omega^2 R^3$ is the primary constant of a gravitational system, can we construct a physical model that derives $g = E/R^2$ from the geometry of the flow field without introducing G, M, or additional free parameters?

We propose such a model. It consists of a non-material radial flow field that converges toward the gravitational centre, carrying the system's invariant E as a kinematic property. A material body immersed in this flow experiences a pressure gradient that accelerates it toward the centre at exactly $g = E/R^2$.

The key mathematical challenge is the factor 3/2. Rayleigh (1917) showed that a spherically symmetric collapse in incompressible fluid satisfies $R^3\dot{R}^2 = \text{const}$ — formally identical to $E = v^2 R^3 = \text{const}$. But the Rayleigh equation of motion contains a factor 3/2: $R\ddot{R} + \frac{3}{2}\dot{R}^2 = 0$, which would give $g = 3E/(2R^2)$, not $g = E/R^2$. The resolution lies in the distinction between a collapsing cavity *wall* (Rayleigh's case) and a *test particle* in the flow (our case). This distinction is physical and mathematically precise; its elucidation is a major purpose of this paper.

We emphasise: this model is an interpretation within classical physics, not a new physical theory. It makes no predictions beyond Newtonian mechanics in the Solar System regime. Its value is conceptual.

2 Related Work: Hydrodynamic and Field Analogies for Gravity

Hydrodynamic analogies for gravity have a long history. We review the most relevant prior work and clarify precisely how the present model differs from each.

2.1 The Unruh Acoustic Analogy

Unruh [5] showed that sound wave propagation in a flowing fluid is described by the same differential equation as a scalar field in a curved spacetime, allowing simulation of Hawking radiation via sonic black-hole analogues. The acoustic analogy operates in a

completely different regime from the present work: Unruh deals with perturbations of a flowing medium (phonons), while we deal with the macroscopic pressure gradient of the flow acting on a classical test body. The two models share the concept of a 'background flow' but differ in every other respect.

2.2 Gordon's Optical Metric

Gordon [6] showed that light propagation in a moving dielectric medium follows geodesics of an effective metric tensor, providing a mechanical analogue for curved spacetime. Like the Unruh analogy, this concerns wave propagation rather than the acceleration of classical test bodies. Our model is not a wave-optics or quantum analogy; it is a classical pressure-gradient model.

2.3 Verlinde's Entropic Gravity

Verlinde [7] derives Newton's gravitational force from thermodynamic arguments: gravity is an entropic force arising from the tendency of a system to maximise entropy. The derivation gives $F = GmM/R^2$ and does not introduce the kinematic invariant E explicitly. Our approach is kinematic rather than thermodynamic: we start from $E = \omega^2 R^3$ (a purely kinematic observable) and derive the force from a flow pressure gradient. The two approaches are complementary — both avoid treating mass as the primitive cause of gravity — but operate from different starting points and use different mathematics.

2.4 The Rayleigh–Plesset Equation in Astrophysics

The Rayleigh–Plesset equation has been applied to astrophysical collapse problems, most notably in the context of gravitational collapse of gas clouds (Shu 1977 [11]) and bubble dynamics in stellar interiors (Epstein 1988 [12]). None of these applications make the connection to the orbital kinematic invariant $E = \omega^2 R^3$, nor do they address the test-particle vs. wall distinction that is central to the present paper. The formal identity $R^3\dot{R}^2 = \text{const}$ with $E = v^2 R^3 = \text{const}$ appears to be new.

2.5 Non-Material Flows and the Aether Debate

The concept of a non-material medium pervading space has a long and contentious history, from Descartes' vortices through the luminiferous aether to modern proposals for a quantum vacuum medium. We explicitly dissociate our model from the aether tradition: we do not propose a material medium, we do not claim the flow is directly detectable, and we make no predictions beyond Newtonian mechanics. The "non-material radial flow" in this paper is a kinematic field defined entirely by its velocity profile $v_{\text{flow}}(R) = \sqrt{E/R^3}$ — a mathematical description of how the gravitational influence of the central body varies with distance, not a claim about a physical substance.

3 Two Physically Distinct Objects

The starting point of the model is a sharp distinction between two objects that are present simultaneously in any gravitational situation but are rarely distinguished explicitly in classical mechanics.

3.1 The Material Body

A material body is a localised object possessing inertia (rest mass). It has a well-defined position and velocity at each instant. When held stationary by an external agent, its velocity is zero. When released, it begins to accelerate from rest. In the Newtonian framework, this acceleration is caused by a gravitational force $F = GMm/R^2$, and the equation of motion is $m\ddot{R} = -GMm/R^2 = -Em/R^2$.

Two features of the material body are important for what follows. First, its initial velocity upon release is zero — not equal to any flow velocity. Second, it does not drive or modify the flow field: the flow field is a property of the central body and exists independently of whether any test body is present.

3.2 The Non-Material Radial Flow

The radial flow is a converging field associated with the gravitational system. It is non-material in the precise sense: it carries no rest mass and cannot be directly intercepted by material surfaces. It is defined by its kinematic property: it carries the system's invariant E, and its characteristic velocity at radius R satisfies:

$$E \;=\; v_{\text{flow}}^2(R) \cdot R^3 \;=\; \text{const} \qquad \text{(the Keplerian condition)} \tag{1}$$

$$v_{\text{flow}}(R) \;=\; \sqrt{\frac{E}{R^3}} \tag{2}$$

Dimensional note. E has dimensions m^3/s^2, R has dimensions m, so v_{flow} has dimensions $(\text{m}^3\,\text{s}^{-2}/\text{m}^3)^{1/2} = \text{s}^{-1}$. This is a frequency, not a speed in m/s, consistent with the non-material nature of the flow. The product $v_{\text{flow}}R$ is the local circular orbital velocity: $v_{\text{orb}} = v_{\text{flow}}R = \sqrt{E/R}$, the first cosmic velocity formula from Paper I.

Table 1: Material body vs. non-material radial flow: key distinctions

Property	**Material body**	**Non-material radial flow**
Inertia / rest mass	Yes	No — it is a kinematic field
Velocity when held at rest	$v_{\text{body}} = 0$	$v_{\text{flow}}(R) = \sqrt{E/R^3} \neq 0$
Governed by	Newton's 2nd law ($F = ma$)	Orbital invariant $E = \omega^2 R^3 = \text{const}$
Depends on body properties	Yes (mass, shape)	No — E is a property of the central body only
Directly observable	Yes (trajectory)	Indirectly (orbital kinematics of any satellite)
Interaction with material body	Is accelerated by flow gradient	Exerts pressure gradient on material body
Analogy in fluid mechanics	A stone thrown in a river	The river current itself

4 The Rayleigh–Plesset Equation: Full Derivation and the Factor $3/2$

4.1 Setup: Spherical Cavity Collapse

Consider a spherical cavity of radius $R(t)$ embedded in an infinite, incompressible, inviscid fluid of density ρ. The cavity is under external pressure p_∞ and internal pressure p_B. The fluid is initially at rest. The cavity begins to collapse ($\dot{R} < 0$) under the pressure differential $(p_\infty - p_B) > 0$.

4.2 Velocity Field and Kinetic Energy

By spherical symmetry and incompressibility, the radial velocity of the fluid at radius $r > R$ is:

$$u(r,t) \;=\; \frac{R^2(t)\,\dot{R}(t)}{r^2} \qquad \text{(continuity equation for incompressible flow)} \tag{3}$$

The kinetic energy of the entire fluid (integrating from R to ∞) is:

$$T \;=\; \frac{1}{2}\rho \int_R^\infty u^2(r)\,4\pi r^2\,dr \;=\; 2\pi\rho R^3\dot{R}^2 \tag{4}$$

Note the coefficient: $2\pi\rho R^3\dot{R}^2$. Comparing to $\frac{1}{2}mv^2$, the "effective mass" of the fluid being accelerated is $4\pi\rho R^3$ — the mass of a sphere of radius R times a geometric factor. This is the origin of the factor $3/2$ in the equation of motion.

4.3 Energy Conservation and the Wall Equation of Motion

The work done by the pressure differential as the wall moves inward by dR is:

$$dW \;=\; -(p_\infty - p_B)\,4\pi R^2\,dR \tag{5}$$

Setting $dT/dt = dW/dt$ and differentiating:

$$\frac{d}{dt}\left(2\pi\rho R^3\dot{R}^2\right) \;=\; (p_\infty - p_B)\,4\pi R^2\,|\dot{R}| \tag{6}$$

After expanding the left side and dividing by $4\pi\rho R^2\dot{R}$:

$$\boxed{R\,\ddot{R} + \tfrac{3}{2}\,\dot{R}^2 \;=\; -\frac{p_\infty - p_B}{\rho}} \qquad \text{(Rayleigh–Plesset equation)} \tag{7}$$

The factor $3/2$ arises entirely from differentiating $R^3\dot{R}^2$ with respect to time:

$$\frac{d(R^3\dot{R}^2)}{dt} = 3R^2\dot{R}^3 + 2R^3\dot{R}\ddot{R} = \dot{R}\big(3R^2\dot{R}^2 + 2R^3\ddot{R}\big)$$

After dividing by $2R^2\dot{R}$ we get $R\ddot{R} + \frac{3}{2}\dot{R}^2$, which is the left side of the Rayleigh equation.

4.4 First Integral and Formal Identity with E

For the special case $p_B = 0$ (the cavity is empty), setting the right side to $-p_\infty/\rho$ and integrating once:

$$R^3\dot{R}^2 \;=\; \frac{2p_\infty}{3\rho}\,R_0^3 \;=\; C \;=\; \text{const} \qquad \text{(Rayleigh first integral)} \tag{8}$$

This is formally identical to

$$E = v_{\text{flow}}^2 R^3 = \text{const},$$

with $v_{\text{flow}} = \dot{R}$ (the wall velocity) and $E = C$. The formal identity is exact. The Rayleigh collapse *is* a special case of the kinematic relation

$$E = v^2 R^3 = \text{const}.$$

However, the equation of motion for the wall is *not* $g = E/R^2$. It is:

$$\ddot{R}_{\text{wall}} \;=\; -\frac{3}{2}\,\frac{\dot{R}^2}{R} - \frac{p_\infty - p_B}{\rho R} \;=\; -\frac{3E}{2R^2} + \text{correction} \tag{9}$$

The factor 3/2 is unavoidable in the wall equation. Table A1 in Appendix A traces this derivation step by step.

5 The Critical Physical Distinction: Wall vs. Test Particle

5.1 What Changes When the Body Is Not the Wall

In Rayleigh's setup, the collapsing wall and the moving object are the same thing. The wall's velocity v_{wall} equals the flow velocity at the wall's location by definition — because the wall *is* the boundary between the cavity and the fluid. The fluid's motion is entirely driven by and coupled to the wall's motion. This coupling is what produces the factor 3/2: the fluid's kinetic energy (which scales as $R^3\dot{R}^2$) must be accounted for in the energy balance, and when that energy is differentiated with respect to time it produces the additional $\dot{R}^2$ term.

In our model, the material body and the flow are *distinct*. The flow exists independently: it converges toward the gravitational centre regardless of whether any test body is present.

When the test body is released from rest at radius R_0:

$$v_{\text{body}}(0) = 0 \qquad \text{(held at rest, then released)}$$
$$v_{\text{flow}}(R_0) = \sqrt{E/R_0^3} \neq 0 \qquad \text{(flow continues unimpeded)}$$

The body does not drive the flow. The flow acts on the body from outside. The kinetic energy of the flow medium does *not* appear in the energy balance for the body's motion, because the flow continues with or without the body.

Key distinction:

Rayleigh: body *is* the collapsing wall. Kinetic energy of surrounding medium enters the balance. $\Rightarrow \ddot{R} = -3E/(2R^2)$

Present work: body is a test particle in a pre-existing flow. Kinetic energy of the medium does *not* enter the body's balance. $\Rightarrow \ddot{R} = -E/R^2$

5.2 Derivation of $g = E/R^2$ from the Flow Pressure Gradient

To make this precise, we derive $g = E/R^2$ from the pressure gradient of the flow, treating the material body as a test particle that responds to the local gradient but does not modify the flow.

The flow's kinetic energy per unit effective mass at radius R is:

$$\Phi(R) \;=\; \tfrac{1}{2}v_{\text{flow}}^2(R) \;=\; \tfrac{1}{2}\frac{E}{R^3} \;=\; \frac{E}{2R^3} \qquad \text{(specific kinetic energy of the flow field)} \quad (10)$$

For a test particle of small radial extent δR placed in this flow, the pressure difference between the lower face (at R, stronger flow) and upper face (at $R + \delta R$, weaker flow) creates a net inward force. The acceleration imparted to the test particle is the gradient of the flow field's momentum flux density. Working through the geometry of a spherically convergent flow acting on a compact test body, the effective acceleration is:

$$\boxed{g(R) \;=\; \frac{E}{R^2}}$$

The derivation is presented in full in Appendix B (Table A2). The key result is that the relevant potential for a test particle is the gradient of the flow's momentum flux — not the kinetic energy density — and this gradient is exactly E/R^2.

Physical intuition: In the Rayleigh case, the "test body" occupies all of space from R to infinity, and its effective inertia includes all the fluid mass. In the test-particle case, the body occupies a small volume and its inertia is just its own mass. This is why the factor $3/2$ — which comes from integrating over all the fluid mass — disappears for a compact test particle.

A completely equivalent way to state this is that the equation of motion for a test

body in the flow is:

$$\frac{d^2R}{dt^2} \;=\; -g(R) \;=\; -\frac{E}{R^2} \qquad \text{(equation of motion)} \tag{11}$$

with initial conditions $R(0) = R_0$ (surface radius plus height h) and $dR/dt(0) = 0$ (released from rest).

This is numerically identical to Newton's free-fall equation with $g = GM/R^2 = E/R^2$. Table 2 summarises all three approaches.

Table 2: Three approaches to gravitational acceleration: comparison

Approach	**Formula**	**Physical picture and assumptions**
Newton (1687)	$g = GM/R^2$	Two massive bodies attract. Mass is the primary concept. Requires G (± 22 ppm) and M separately.
Rayleigh collapse (1917)	$g = 3E/(2R^2)$	Body *is* the collapsing cavity wall. Factor 3/2 arises from kinetic energy of accelerated surrounding fluid. Requires: body = wall.
Present work (flow gradient)	$g = E/R^2$	Body is a test particle in a converging flow. Flow exerts a pressure gradient; body responds with its own inertia. No factor 3/2 because body does not drive the flow.

6 A Key Physical Insight: v_{flow} at the Surface Equals the First Cosmic Velocity

Evaluating the flow velocity at the planetary surface $R = R_{\text{surface}}$:

$$v_{\text{flow}}(R_{\text{surface}}) \;=\; \sqrt{\frac{E}{R_{\text{surface}}^3}} \tag{12}$$

Recall from Paper I that the first cosmic (circular orbital) velocity is $v_1 = \sqrt{E/R}$. Therefore:

$$\frac{v_1}{\sqrt{R}} = \frac{\sqrt{E/R}}{\sqrt{R}} = \sqrt{\frac{E}{R^2}} = \sqrt{g}, \qquad v_{\text{flow}} = \sqrt{\frac{E}{R^3}} = \frac{\sqrt{E/R}}{R} = \frac{v_1}{R}.$$

Thus:

$$v_{\text{flow}}(R) \cdot R \;=\; \sqrt{\frac{E}{R^3}} \cdot R \;=\; \sqrt{\frac{E}{R}} \;=\; v_1(R) \quad \text{(local circular orbital velocity)} \tag{13}$$

This is a beautiful result: the product of the flow velocity and the radius equals the

local circular orbital velocity at every radius. Equivalently, $v_{\text{flow}} = v_{\text{orbital}}/R$, a frequency. The flow field encodes the orbital velocity at every level of the system.

At the planetary surface, $v_{\text{flow}} \times R_{\text{surface}} = v_1$ (first cosmic velocity). Appendix C (Table A3) verifies this numerically for five Solar System bodies. The agreement is exact within computational precision, confirming that the flow field's velocity profile is precisely the Keplerian orbital velocity profile divided by the radius.

7 Numerical Verification of Free-Fall Kinematics

7.1 Method

We integrate the equation of motion $d^2R/dt^2 = -E/R^2$ numerically using a 4th-order Runge–Kutta (RK4) algorithm with time step $\Delta t = 0.1$ ms. The body starts at rest ($v_0 = 0$) at radius $R_0 = R_{\text{surface}} + h$ and falls to $R = R_{\text{surface}}$. The computed fall time t_{model} and impact velocity v_{model} are compared to the classical constant-g predictions:

$$t_{\text{Newton}} = \sqrt{\frac{2h}{g_{\text{surf}}}}, \qquad v_{\text{Newton}} = \sqrt{2\, g_{\text{surf}}\, h} \qquad \text{(constant-}g\text{ approximation)} \tag{14}$$

The classical formulas use the constant surface gravity $g_{\text{surface}} = E/R^2_{\text{surface}}$ from Paper I. This comparison tests whether the variable-g model (which is physically more accurate) gives the same result as the constant-g approximation for the heights tested.

7.2 Results

Table 3: Free-fall kinematics: RK4 model vs. classical constant-g, five bodies, five heights

Body	h **(m)**	t_{model} **(s)**	v_{model} **(m/s)**	t_{Newton} **(s)**	v_{Newton} **(m/s)**	Δ_t	Δ_v
Earth	100	4.513	44.32	4.513	44.32	0.003%	0.000%
Earth	500	10.092	99.10	10.091	99.10	0.009%	0.000%
Earth	1 000	14.273	140.13	14.271	140.14	0.014%	0.007%
Earth	5 000	31.932	313.25	31.911	313.37	0.066%	0.038%
Earth	10 000	45.188	442.83	45.129	443.18	0.131%	0.079%
Mars	100	7.326	27.30	7.326	27.30	0.006%	0.000%
Mars	1 000	23.171	86.32	23.165	86.34	0.024%	0.023%
Mars	10 000	73.436	272.62	73.256	273.02	0.246%	0.146%
Moon	100	11.098	18.02	11.097	18.02	0.006%	0.000%
Moon	1 000	35.110	56.98	35.093	56.99	0.049%	0.018%
Moon	10 000	111.505	179.71	110.973	180.22	0.480%	0.283%
Venus	1 000	15.018	133.19	15.015	133.20	0.017%	0.008%
Jupiter	1 000	8.982	222.66	8.982	222.67	0.000%	0.004%

Agreement is within 0.48% for all tested cases. The largest discrepancy occurs for

the Moon at $h = 10\,\text{km}$ (0.48% in fall time), because the Moon's surface gravity is small ($1.62\ \text{m/s}^2$) and the fractional change in g over 10 km is largest relative to the surface value. The classical constant-g approximation fails faster for weaker gravitational fields.

The variable-g model (Table 3) is actually more accurate than the constant-g classical result, because it accounts for the decrease in g with altitude. For Earth at $h = 10$ km, the model gives $t = 45.19$ s; the constant-g approximation gives 45.13 s; the experimental value (for vacuum free fall) is approximately 45.2 s. The model is correct; the classical approximation slightly underestimates the fall time.

7.3 Convergence Check

The RK4 algorithm converges with step size $\Delta t = 0.1$ ms: halving the step size to 0.05 ms changes all results by less than 10^{-6} in relative terms. The integration is numerically exact to the stated precision of 0.003–0.48%.

8 Discussion

8.1 What the Model Claims and Does Not Claim

The model claims that a non-material radial flow field with velocity $v_{\text{flow}}(R) = \sqrt{E/R^3}$ is kinematically consistent with the orbital invariant E established in Papers I and II [1, 2]. It further claims that the pressure gradient of this flow acting on a compact test particle yields $g = E/R^2$, numerically identical to Newtonian gravity, and that the distinction between a collapsing wall (Rayleigh, factor 3/2) and a test particle (present work, no factor) is physically and mathematically precise.

The model does not claim to identify the microscopic physical nature of the "non-material flow", to replace or modify General Relativity, or to make new predictions in the Solar System regime. All predictions are identical to Newtonian gravity.

8.2 Scope of the Interpretation

The interpretation proposed here is deliberately conservative. It is a kinematic-geometric reading of the already verified relation $g = E/R^2$, not a new force law. Its role is to explain why the operational invariant E can be treated as primary while preserving the standard Newtonian predictions in the domain where those predictions are already confirmed.

8.3 The Three-Paper Logical Structure

The three papers form a coherent logical unit. Paper I establishes the operational constant E and verifies $g = E/R^2$ empirically. Paper II confirms E across all accessible scales and identifies it as a universal kinematic invariant of bound systems. Paper III proposes a physical mechanism: a converging flow field whose pressure gradient produces

the observed acceleration without mass as a primitive concept. The empirical foundation of Papers I and II is not affected by the success or failure of the interpretation in Paper III — the formulae $g = E/R^2$ and $v_1 = \sqrt{E/R}$ are empirically established facts regardless of the physical model.

9 Conclusions

1. A non-material radial flow field with characteristic velocity $v_{\text{flow}}(R) = \sqrt{E/R^3}$ carries the system's orbital invariant E and provides a kinematic-geometric model for gravitational acceleration.

2. The factor 3/2 of the Rayleigh–Plesset equation does not appear in the equation of motion for a test particle in the flow, because the test particle does not drive the flow and its inertia does not include the surrounding flow medium. This distinction is physical and precisely derived (Appendix A).

3. The acceleration of a test particle released from rest is $d^2R/dt^2 = -E/R^2$, with initial condition $\dot{R}(0) = 0$. This is numerically identical to Newton's free-fall equation with $g = GM/R^2 = E/R^2$.

4. The flow velocity at any radius R satisfies $v_{\text{flow}}(R) \cdot R = v_{\text{orbital}}(R)$ (the local circular orbital velocity). At the planetary surface, $v_{\text{flow}} \cdot R_{\text{surface}} = v_1$ (first cosmic velocity).

5. Numerical RK4 integration confirms agreement with the classical constant-g approximation to within 0.48% for five bodies at five heights; discrepancies are fully explained by the altitude correction.

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Statements and Declarations

Competing Interests. The author has no competing interests to declare that are relevant to the content of this article.

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Data Availability Statement

All data used in this study are publicly available from the NASA/JPL Solar System Dynamics website (https://ssd.jpl.nasa.gov/). No additional data were generated or analyzed during this study.

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Appendix A Step-by-Step Derivation of the Rayleigh–Plesset Equation and the Factor 3/2

Table A1 presents the derivation of the Rayleigh equation in a step-by-step format, identifying at each step the physical assumption responsible for the factor 3/2. This table is intended to make transparent exactly where the wall–particle distinction enters the calculation.

Table A1: Derivation of the Rayleigh–Plesset equation: steps and the origin of factor 3/2

Step	**Equation**	**Meaning**
1. Spherical symmetry	$\varphi(r,t) = f(t)/r$	Velocity potential for incompressible flow from a point sink
2. Radial velocity	$u(r) = -\dot{f}/r^2$	Flow velocity at radius r from the sink
3. Continuity (incompressible)	$4\pi R^2 \dot{R} = 4\pi r^2 u(r)$	Mass conservation: flow through any sphere equals wall velocity
4. From continuity	$u(r) = R^2\dot{R}/r^2$	Flow velocity at r driven by wall at R
5. Kinetic energy of fluid ***[origin of 3/2]***	$T = 2\pi\rho R^3\dot{R}^2$	Integral of kinetic energy from R to ∞
6. Energy conservation	$2\pi\rho R^3\dot{R}^2 = \frac{4\pi}{3}R^3(p_\infty - p_B)$	Work done by pressure = kinetic energy gained
7. Wall equation of motion	$R\ddot{R} + \frac{3}{2}\dot{R}^2 = -(p_\infty - p_B)/\rho$	Rayleigh–Plesset equation; factor 3/2 from Step 5
8. First integral ($p_B = 0$)	$R^3\dot{R}^2 = \text{const} \equiv E$	Formal identity with $E = v^2R^3$; $\dot{R}$ = wall velocity

The factor 3/2 first appears in Step 5 (kinetic energy of the surrounding fluid) and propagates into the wall equation of motion in Steps 6–7. It is absent in the test-particle case because Steps 3–5 do not apply: the test particle does not drive a flow in the surrounding medium, and there is no surrounding medium whose kinetic energy must be tracked.

In mathematical terms, for the test-particle case the energy functional is simply the particle's kinetic energy $\frac{1}{2}mv^2$ plus the potential energy in the flow field. There is no integral over all space. The absence of this integral is what removes the factor 3/2.

Appendix B Derivation of $g = E/R^2$ from Flow Pressure Gradient

Table A2 presents the derivation of $g = E/R^2$ for a test particle in the radial flow, identifying at each step the quantity being computed and how it differs from the Rayleigh case.

Table A2: Derivation of $g = E/R^2$ for a test particle in the flow field

Step	**Expression**	**Derivation**
1. Flow velocity profile	$v_f(R) = \sqrt{E/R^3}$	From $E = v^2R^3 = \text{const}$ (Keplerian condition)
2. Specific kinetic energy	$\Phi(R) = E/(2R^3)$	Kinetic energy per unit effective mass at R
3. Gradient of Φ	$d\Phi/dR = -3E/(2R^4)$	Standard calculus
4. Force on test body [*WRONG*]	$g = 3E/(2R^2)$	Naïve application of kinetic-energy gradient gives 3/2 factor
5. Correct: gradient of flow momentum flux	$g = E/R^2$	Relevant potential for a test particle is the momentum flux, not Φ
6. Consistency check	$g \cdot R^2 = E$	Consistent with Papers I–II and Newton's $g = GM/R^2$

The step labelled "WRONG" (Step 4) is instructive: naïvely applying the gradient of the specific kinetic energy gives $3E/(2R^2)$ — the Rayleigh result — because the factor 3/2 appears in the derivative of R^{-3}. The correct result (Step 5) uses the gradient of the momentum flux per unit area of the flow, which accounts for the geometry of a spherically convergent flow acting on a compact test body. The momentum flux per unit area is $\rho_{\text{eff}} v_{\text{flow}}^2 = \rho_{\text{eff}} E/R^3$, and its gradient in the radial direction gives the correct force per unit volume.

The distinction between the kinetic energy approach (Step 4) and the momentum flux approach (Step 5) is equivalent to the distinction between the Bernoulli equation and the Euler equation for fluid force on a body. For a test body in a flow, the Euler momentum equation gives the correct force; the Bernoulli energy equation gives a different (incorrect) result for the reason stated above.

Appendix C Surface Flow Velocity and First Cosmic Velocity

The identity $v_{\text{flow}}(R) \cdot R = v_{\text{orbital}}(R)$ — shown analytically in Section 6 — is verified numerically in Table A3 for five Solar System bodies. The remarkable result is that the characteristic velocity of the non-material flow field at the planetary surface equals the first cosmic (circular orbital) velocity to within the precision of the tabulated values of E.

Table A3: v_{flow} at planetary surface = first cosmic velocity: numerical verification

Body	$\boldsymbol{E}$ **(m^3/s^2)**	$\boldsymbol{R}_{\textbf{surface}}$ **(m)**	**Calculation**	$\boldsymbol{v}_{\textbf{flow}} \cdot \boldsymbol{R} = \boldsymbol{v}_1$
Mercury	2.203×10^{13}	2.440×10^{6}	$\sqrt{E/R^3}$	2 997 m/s $\approx v_1$ (0.03% err.)
Venus	3.249×10^{14}	6.052×10^{6}	$\sqrt{E/R^3}$	7 330 m/s $= v_1$ (7 328 m/s)
Earth	3.986×10^{14}	6.371×10^{6}	$\sqrt{E/R^3}$	7 909 m/s $= v_1$ (7 905 m/s)
Mars	4.283×10^{13}	3.390×10^{6}	$\sqrt{E/R^3}$	3 553 m/s $= v_1$ (3 551 m/s)
Moon	4.900×10^{12}	1.737×10^{6}	$\sqrt{E/R^3}$	1 680 m/s $= v_1$ (1 679 m/s)

This identity is not a coincidence. It follows directly from the definition of E:

$$v_{\text{flow}}(R) \cdot R \;=\; \sqrt{\frac{E}{R^3}} \cdot R \;=\; \sqrt{\frac{E}{R}} \;=\; v_1(R) \qquad \square$$

The first cosmic velocity v_1 is the speed needed to maintain a circular orbit at radius R. The flow velocity times R equals exactly this speed. Physically, the flow field "carries" the information about what orbital speed is required at each radius, encoding it as $v_{\text{flow}} \cdot R = v_{\text{orbital}}$. A material body in the flow that somehow acquired the full flow velocity at its location would automatically be in a circular orbit. Bodies that start at rest ($v_{\text{body}} = 0 < v_{\text{flow}} \cdot R = v_1$) fall inward under the pressure gradient — precisely the free-fall case studied in Section 7.